

Directional Derivative and the Gradient Vector

Trim Ch 11, S. 12.8, 12.4

Outline:

1. Directional derivative and the gradient vector
2. Properties of the directional derivative and the gradient vector

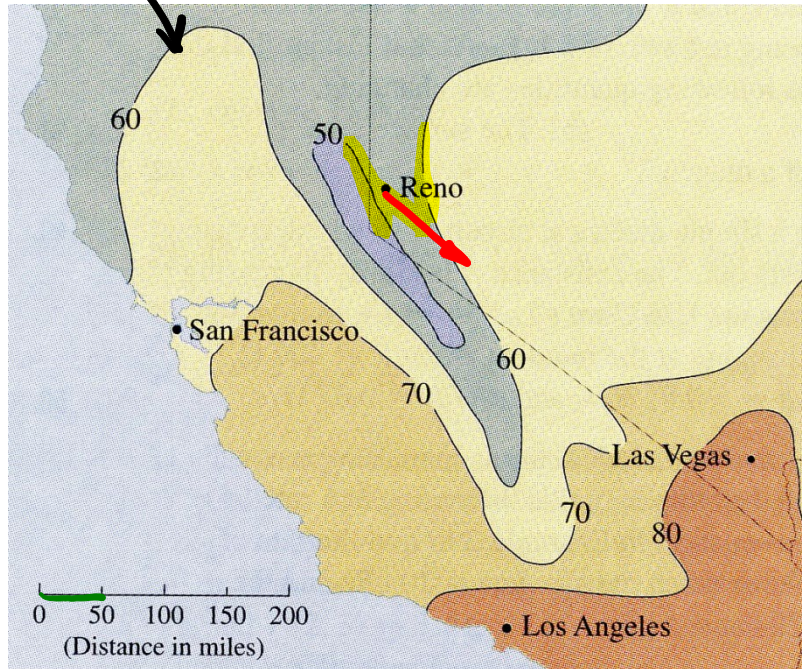


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Weather Maps

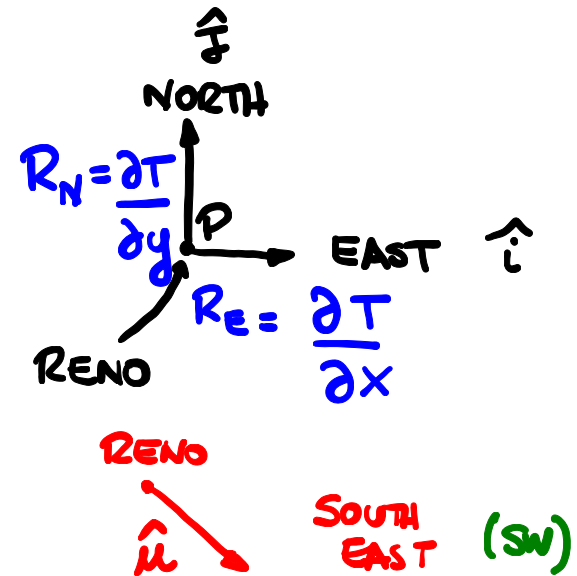
ISOTHERMS

Level curves (or isotherms) join locations with the same temperature



Contour map of the temperature function $T(x,y)$ for the states of California and Nevada at 3:00pm on October 10, 1997.

At a location such as Reno, what if we want to know the rate of change of temperature when traveling southeast?



$$R_{sw} = \frac{60 - 50}{75} \frac{^{\circ}\text{F}}{\text{miles}} \approx 0.13 \frac{^{\circ}\text{F}}{\text{miles}}$$

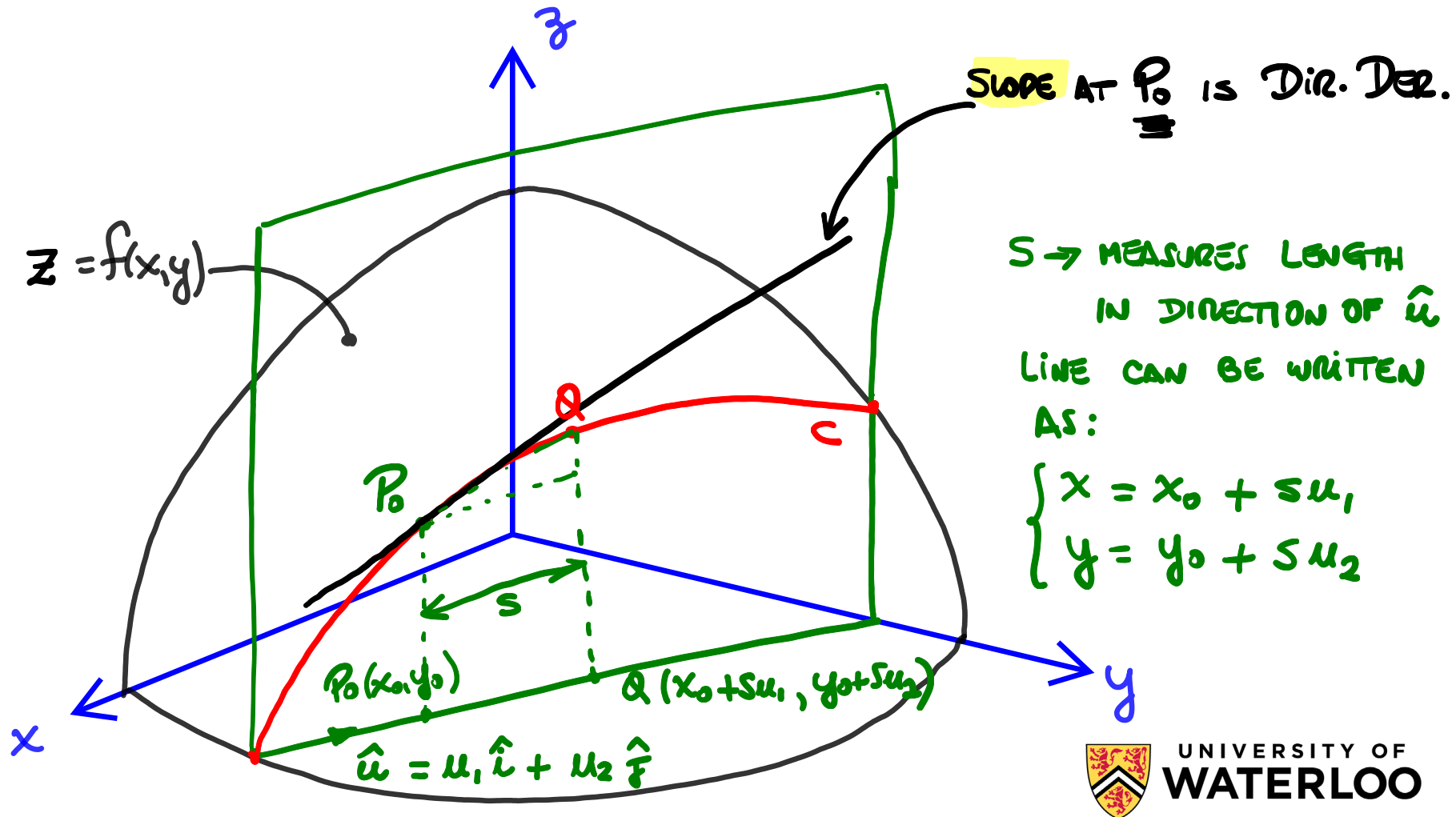
DIRECTIONAL DERIVATIVE $\hat{u}$ TO LAS VEGAS $T(x,y)$



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Directional Derivative

Given the surface $f(x, y)$, suppose we want to step off in the direction of an arbitrary vector $\hat{u} = (u_1, u_2)$ of increasing “ s ”



Directional Derivative

If the parameter “s” measures the length of the line in the direction of $\hat{u}$, the equation of this line can be written as,

$$\begin{cases} x = x_0 + s u_1 \\ y = y_0 + s u_2 \end{cases}$$

As the point Q gets closer to the point P_0 , we can calculate the slope of the tangent line (or rate of change) of $f(x, y)$ at P_0 stepping off in the direction of $\hat{u}$ as,

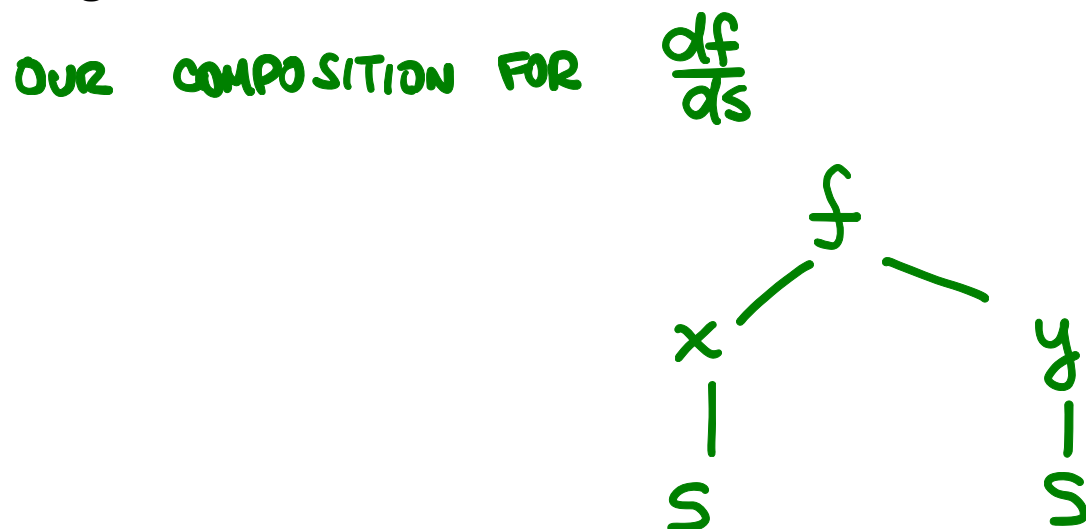
$$\left(\frac{df}{ds} \right)_{\hat{u}, P_0} = \frac{f(x_0 + s u_1, y_0 + s u_2) - f(x_0, y_0)}{s}$$

In the limit, this is called the Directional derivative of $f(x, y)$ at P_0 in the direction of $\hat{u}$.



Directional Derivative and the Gradient Vector

Let us now develop an efficient formula to calculate the directional derivative using the chain rule,



APPLYING CHAIN RULE,

$$\left(\frac{df}{ds}\right)_{\hat{u}, P} = \left(\frac{\partial f}{\partial x}\right)_P \cdot \underbrace{\frac{dx}{ds}}_{\mu_1} + \left(\frac{\partial f}{\partial y}\right)_P \cdot \underbrace{\frac{dy}{ds}}_{\mu_2}$$



Directional Derivative and the Gradient Vector

The directional derivative of a function $f(x, y)$ at P_0 , stepping off in the direction of $\hat{u}$ can therefore be calculated from,

$$\left(\frac{df}{ds}\right)_{\hat{u}, P_0} = \underbrace{\left[\left(\frac{\partial f}{\partial x}\right)_{P_0} \hat{i} + \left(\frac{\partial f}{\partial y}\right)_{P_0} \hat{j} \right]}_{\text{GRADIENT VECTOR [12.4]}} \cdot \underbrace{\left[u_1 \hat{i} + u_2 \hat{j} \right]}_{\hat{u}}$$

where,

$$\vec{\nabla} f = \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} \right)$$

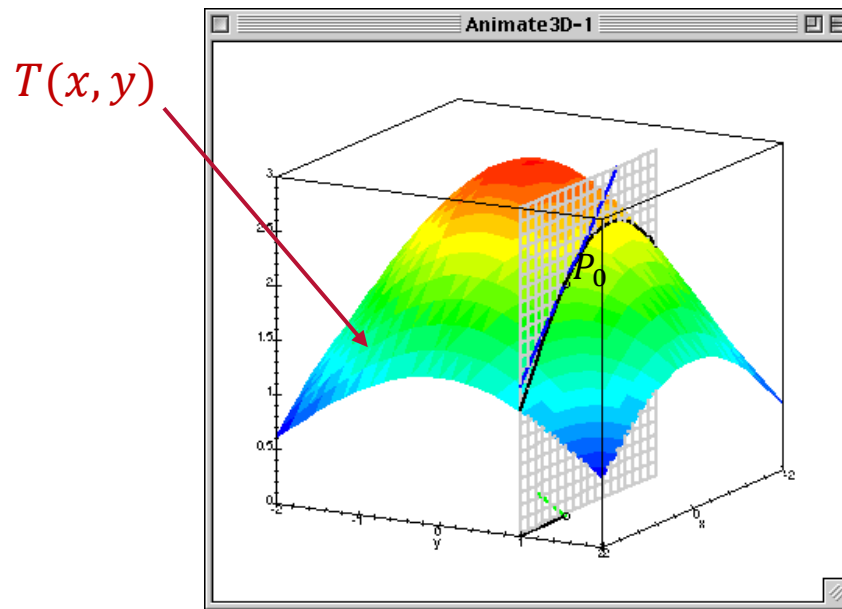
is called the gradient vector. **AND** .

$$\left(\frac{df}{ds}\right)_{\hat{u}, P_0} = \vec{\nabla} f \big|_{P_0} \cdot \hat{u} = \left(\nabla_{\hat{u}} f \right)_{P_0}$$



Directional Derivative – Physical Interpretation

If $T(x, y)$ is the temperature at each point $P_0(x_0, y_0)$ of the surface, defined over a region R ,



The directional derivative is the **instantaneous rate of change** of temperature at P_0 **stepping off in the direction of $\hat{u}$**

$$D_{\hat{u}}T|_{P_0} = \left. \frac{dT}{ds} \right|_{\hat{u}, P_0} = \vec{\nabla}T|_{P_0} \cdot \hat{u}$$

Example:

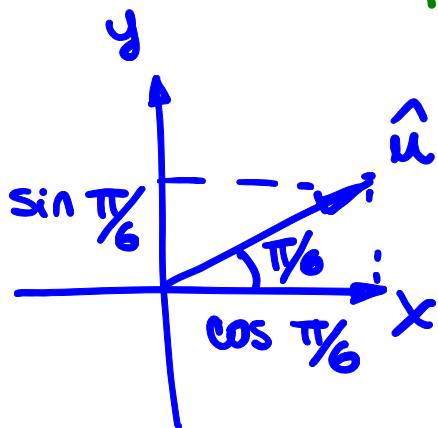
Find the directional derivative $D_{\hat{u}}f(x, y)$ at $(1, 2)$ of the function,

$$f(x, y) = x^3 - 3xy + 4y^2$$

where $\hat{u}$ is the unit vector given by the angle $\theta = \frac{\pi}{6}$

Solution:

$$D_{\hat{u}}f(x, y) \Big|_{(1, 2)} = \left(\vec{\nabla} f \right)_{P_0} \cdot \hat{u} = ?$$



$$\hat{u} = \left(\cos \frac{\pi}{6}, \sin \frac{\pi}{6} \right) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right)$$



Example:

$$\vec{\nabla}_f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} = (3x^2 - 3y, -3x + 8y)$$

$$\begin{aligned} \vec{\nabla}_f \big|_{(1,2)} &= ((3(1)^2 - 3(2)), (-3(1) + 8(2))) \\ &= (-3, 13) \end{aligned}$$

$$\begin{aligned} D_{\hat{u}} f \big|_{(1,2)} &= (-3, 13) \cdot \left(\frac{\sqrt{3}}{2}, \frac{1}{2} \right) \\ &= \frac{-3\sqrt{3} + 13}{2} \end{aligned}$$

← SCALAR ! !



Properties of the Directional Derivative and the Gradient Vector

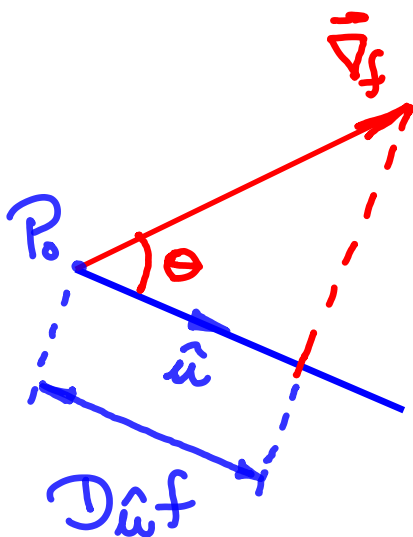


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Directional Derivative in the Direction of $\hat{u}$

The directional derivative of a function $f(x, y)$ at P_0 in the direction of $\hat{u}$ gives us insight into some of the properties of the gradient vector,

$$D_{\hat{u}}f|_{P_0} = \left. \frac{df}{ds} \right|_{\hat{u}, P_0} = \vec{\nabla}f|_{P_0} \cdot \hat{u}$$



$$|D_{\hat{u}}f| = |\vec{\nabla}f| |\hat{u}| \cos\theta$$

$\hat{u} = 1$

$$D_{\hat{u}}f = |\vec{\nabla}f| \cos\theta$$

The directional derivative $D_{\hat{u}}f|_{P_0}$ is the component of the gradient vector at P_0 in the direction of $\hat{u}$.



Summary – Part I

- **The directional derivative is the instantaneous rate of change of a function $f(x, y)$ at P stepping off in the direction of $\hat{u}$**
- The directional derivative is the component of the gradient vector in the direction of $\hat{u}$